Abdulrahman Sirelkhatim

Medical Graduate · Health Data Scientist · Organizational Leader

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Professional Summary

First-generation medical graduate (MBBS, SUST, 2026) and health data scientist. Delivered 25 data analysis projects for medical researchers across 10+ specialties, co-authored studies on mass casualty preparedness in Sudan, and led SUMSA (800+ members) as President, all while completing medical school through revolution, COVID-19, a faculty strike and armed conflict. Targeting health data science research and graduate study in computational neuroscience. Over 7 years of progressively responsible roles across medical student organizations, national and international healthcare networks, and a surgical development consortium.

Education

Bachelor of Medicine, Bachelor of Surgery (MBBS) — *Sudan University of Science and Technology* · *11/2018 – 07/2026* · *Khartoum*

- GPA 3.24/4.0 · Ranked 22nd of 127 · First Degree Honors
- Progression maintained through the 2019 revolution, COVID-19, a 2021 faculty strike and the April 2023 armed conflict
- Thesis (Grade A): Knowledge, Attitudes and Practices on Stroke Manifestations and Complications Among Sudanese Population, Khartoum 2025

Certificate in Computer & Data Science — *MIT Emerging Talent Program* · *11/2024 – 12/2025 · 5% acceptance rate*

- Capstone: Humanitarian Crisis Intelligence and Communication System (HCICS), fusing IOM displacement and OpenStreetMap health-facility data to map underserved populations (Python, GeoPandas, Folium, Streamlit)

Pan African Youth Leadership Program — *University of Texas at Austin / US Department of State* · *07/2017*

Research Experience

Trauma Surgeons' Preparedness for Mass Casualty Incidents During the Armed Conflict in Sudan (2025) — *Multicenter qualitative study* · *12/2024 – 04/2025*

- Co-author; interviews with trauma surgeons across five tertiary hospitals in conflict-affected states

Doctors' Perspectives of Hospital Preparedness for Mass Casualty Incidents in Conflict-Affected Regions of Sudan (2025) — *Multicenter qualitative study* · *2025*

- Co-author; thematic analysis of doctors' views on hospital-level preparedness

Knowledge, Attitudes and Practices on Stroke Among Sudanese Population, Khartoum 2025 — *MBBS graduation thesis (Grade A)* · *05/2025 – 09/2025*

- Author; cross-sectional survey (N = 473), descriptive and inferential analysis in Python and SPSS

Um Katti Hospital and Its Catchment Community: Post-Conflict Epidemiological Baseline Assessment — *Rural Health Training Programme, SUST* · *04/2026*

- Main author of the final report; 8-member team; 262-household survey (1,355 persons), 26 key informant interviews, 8 observation checklists; hospital serving about 66,000 people

Evaluating Deep Learning Brain Tumor Classification on African Neuroimaging Data with Grad-CAM and Offline Mobile Deployment — *Independent research* · *Ongoing*

- Main author; proposal stage; domain-shift and explainability evaluation for low-resource settings

Professional Experience

Freelance Health Data Analyst — *Self-employed, Khartoum* · *10/2025 – Present*

- 25 complete data analysis projects for MD, MBBS, Clinical Master and doctoral researchers across 10+ specialties
- Full-service statistics: study design, cleaning, regression, survival analysis, diagnostic test evaluation, publication-ready figures, thesis-defense slides (11 prepared; all clients successfully defended)
- Tools: Python (pandas, NumPy, SciPy, statsmodels, scikit-learn), SPSS, Excel

Deputy Director, Resources & Excellence Management — *Sudanese Consortium for Surgical Development (SCSD)* · *08/2024 – 05/2026*

- Recruitment across 4 departments: 80+ applications reviewed, 20–30 interviews; defined KPIs from Chief Visionary Officer to entry level
- Built the annual monitoring and evaluation framework and the professional development programs; drafted partnership guidelines

Organizer, Fundamentals of Nursing Course — *Sharg Alneel Hospital* · *10/2026 – 12/2026*

- Delivered 3 sessions to nursing staff; created a facilitator portfolio: 43 sessions, 106 hours, 9 universities (2020–2026)

Leadership

President — *SUMSA* · *03/2025 – 12/2025 · 800+ students*

- Rebuilt the association from near-collapse within 2 months; designed the 5-pillar M&E system and a credit-hours system
- MedXplore 2025: 7-day orientation for 465 students; SPDP: 16-session program for 216 members
- Constitutional reform ratified by 400+ voters; ran the 2026 elections with automated tabulation, results in 24 hours

Secretary General (De Facto President) — *SUMSA* · *02/2023 – 12/2023*

- Established a 14,000+ document archive; organized 36 meetings; raised 12,000,000+ SDG for 98+ war-affected students

External Affairs Secretary · Advisory Board Member — *SUMSA* · *2022 – 2023 · 08/2026*

- 371 letters drafted, $14,000+ secured for medical missions; invited to SUMSA's first Advisory Board

SCOPE roles (General Assistant, Capacity Building Assistant, LEO, VP External Affairs) — *MedSIN-Sudan / MedSIN-SUMSA* · *2020 – 2024*

- Founded the TNEF facilitator training (certified facilitators 7 to 20+); drafted SCOPE Sudan's first Capacity Building Guidelines; IFMSA Certified Facilitator (PRET, 2022)
- Total facilitation record: 43 sessions, 106 hours

Field & Clinical

- **Group Head, SUMSA Medical Mission, Garri** (06/2026): led 32 students and a lab team serving 600–650 patients
- **MBBS clinical training** (2022–2026): ten specialties including Internal Medicine, Surgery, OB/GYN, Pediatrics; 132-page rural residency report

Skills, Certifications & Languages

Technical: Python, SPSS, SQL, Excel, Git/GitHub, Google Workspace and Apps Script, machine learning, geospatial analysis
Certificates: IBM Data Science Professional; DeepLearning.AI Mathematics for ML; Google Project Management; MITx 6.00.1x and 6.00.2x; IFMSA Certified Facilitator; IFMSA OGHA Workshop
Languages: Arabic (native); English (fluent, C1, Duolingo 140)

References available upon request.